

Odonus envelopes — a note for the morning

Status: design sketch. AC, evening of 2026-08-07.

"we have envelopes in the FH2 control surface but nothing at all about it in, for example, Odonus which is really where it would be more useful. All we have is length, which probably just surfaces as a long gate"

Correct, and the code agrees. Worth reading before starting, because the work is probably not the work it looks like.

What Odonus has today

Every duration control scales exactly one number — how long the note is held:

```
gateMsFor f = stepMs / max 1.0 spd * (toNumber r.odo.gatePct / 100.0) *  
toNumber f.dur
```

- **f.dur** — the per-cell LENGTH knob
- **gatePct** — a rig-wide gate-length percentage
- **spd** — the head's speed, so a note breathes with its playhead's rate

That is a **rectangle**: on for N milliseconds, off. There is no attack, no decay, no release, and nothing that can differ between the start and the end of a note. Through an FH-2 MCV in gate mode a longer **dur** is literally a longer gate; through an MCV in envelope mode it is a longer *sustain*, which is why it can sound like it is doing something without being an envelope control at all.

The work is routing, not building

Two envelope generators already exist and Odonus reaches neither.

FH-2 MCV envelopes. **envelopeSpec** sets **ve** on a voice, and **set-adsr-ccs** binds that voice's A/D/S/R to four MIDI CCs. There is even a convenience form: **set-envelope-with-ccs v o c** auto-assigns CCs **70 + 4v ... 73 + 4v**, so voice 0 lands on 70–73, voice 1 on 74–77, and so on. The shape becomes drivable live over CC with nothing stored per note.

ES-9 polyenv. A Selene family — an autonomous envelope per output, applied as config rather than played.

So building envelope generation inside Odonus would make a third. Don't. The job is a route to one of the two.

Recommended shape: per PLAYHEAD, not per cell

Odonus has four playheads and sixteen cells. Sixteen cells × four envelope parameters is a control surface nobody would use, and it is not how the hardware thinks either: a voice has an envelope, and the pattern modulates it.

So:

- **The envelope belongs to the head.** Four knobs per playhead — A/D/S/R — written to that head's ADSR CCs. Odonus already sends CCs on a head's channel (it does portamento with CC 65 / CC 5), so the emit path exists; this is knobs plus a mapping, not new plumbing.
- **The cell keeps modulating, not shaping.** **dur** and velocity stay exactly what they are — how long, how hard — and now they modulate a shape rather than being the whole of it. Nothing in the grid has to change.
- **It stays live.** ADSR over CC means the envelope is a performance control you can ride while the sequence runs, which is the point of putting it here rather than in the FH-2 config surface.

That also keeps the artefact honest: a head's envelope is rig-facing state, not part of the pattern, so a saved Odonus pattern does not silently carry a timbre with it. (Same split as today's name-vs-content work: the artefact holds the musical fact, the envelope holds the placement.)

Settle this first: what does "length" mean once there is a release?

The one question worth answering out loud before writing code.

dur currently means gate length, and a gate stops when it stops. An envelope's **release outlives the gate** — so as soon as there is a release stage, "how long is this note" has two answers, and the grid shows one of them. A short cell with a long release sounds long; the LENGTH knob will look like it is lying.

Three things need to agree on the answer — Odonus's **dur**, the FH-2 MCV envelope path, and Balistes' cells (which have their own **durMs**). If each decides separately you will have three notions of duration to reconcile later, and they will disagree only in the interesting cases.

This is the same open language question already recorded as "machines need a trigger-vs-note duration story". Envelopes are what force it.

Possible answers, none obviously right:

1. **dur means gate; release is extra.** Simple, honest, and the grid stops predicting the sound.
2. **dur means total sounding time**, with the gate shortened to compensate. The grid tells the truth, at the cost of the gate no longer being directly settable.
3. **dur means gate, and the head shows its release as a tail on the grid** — the display absorbs the difference rather than the semantics. More work, keeps both facts visible.

Which path first

FH-2 first. It is closer to what Odonus already does: Odonus emits MIDI notes on a head's channel, and MCV envelopes plus ADSR CCs sit directly on that path. ES-9 **polyenv** is autonomous config in the Selene idiom — a different mental model, and it drags in the claims/panic story we spent today fixing (an autonomous envelope keeps running when the transport stops, and only the daemon's **panic** verb stops it).

Rough order

1. Decide the duration semantics above. Cheap on paper, expensive in code.

2. One head, hand-wired: `set-envelope-with-ccs` for its voice, four knobs sending its CCs, and listen. Confirms the CC path end to end before any UI investment.
3. Generalise to four heads.
4. Decide whether envelope settings are per-session rig state or travel with a saved Odonus scene — and if they travel, they are envelope metadata, not part of the pattern.

Where this went

Landed 2026-08-08/09: the FH-2 polyenv path, routed through the unified router (`DESIGN-routing.md`), plus a starter library and per-field keyboard editing in Selene. Two findings worth carrying forward:

- **polyenv wants bipolar5v.depth** is an attenuverter (64 = zero, <64 inverts), so its rest sits at the CENTRE of the output window — only the bipolar window puts that at 0V. On any unipolar range the envelope pulses on a DC pedestal and never closes. Fixed in `familyDefaultRange`.
- **5V is the ceiling** for a single MCV, confirmed against the manual: the usable swing is half the window span. Headroom needs either two MCVs summed on one jack (the output stage is a mixer — UNTESTED) or the ES-9.

The control surface beyond this is `DESIGN-envelope-machine.md` — what to take from Zadar, Tides and Ceis.

Watch for

- `set-envelope-with-ccs`'s **CC block is computed, not configurable** (`70 + 4v`). Four heads is CCs 70–85. Check nothing else on that channel already uses them — the Twister sends CCs 1–16, so no collision there, but the FH-2's own mapping table may.
- **The MCV envelope is per FH-2 voice, and heads are per Odonus playhead.** Those need a stable correspondence, or a head's knobs will shape someone else's note.
- **gatePct is rig-wide.** With envelopes in play it becomes a global that quietly interacts with every head's shape; may want demoting to per-head.